

# Facundo Bacigalupi

## Computer Analyst

+598 096 366 633  
Montevideo, Uruguay  
bacigalupifacundo@gmail.com

LinkedIn: [facundobacigalupi](#)  
GitHub: [FacundoBacigalupi](#)

Computer Analyst and advanced Computer Engineering student at Universidad de la República. I am a software developer with experience in backend, web development, maintenance, and automated testing, mainly working with Java, Spring Boot, Angular, TypeScript, SQL/NoSQL databases, and REST/SOAP services.

I enjoy understanding how systems work from the inside, improving existing codebases, and building practical solutions that are clear, maintainable, and useful for real users. Through both professional experience and personal projects, I have worked across backend services, frontend interfaces, mobile applications, desktop tools, and local automation.

I consider myself curious, persistent, and adaptable. I like learning new technologies by applying them in real projects, solving concrete problems, and improving step by step. I also use generative AI tools as support for debugging, code analysis, technical documentation, and learning, while always reviewing, testing, and adapting the results myself.

### SKILLS

|                     |                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------|
| Languages           | Java, Python, C, C++, JavaScript, TypeScript, SQL, Dart                                         |
| Backend             | Spring Boot, Spring Framework, REST APIs, SOAP, Java EE, Servlets, JSP, JWT, WebSocket          |
| Frontend / Mobile   | React, Angular, Flutter, HTML5, CSS3, Tailwind CSS, Vite                                        |
| Databases           | PostgreSQL, MySQL, MongoDB, SQLite                                                              |
| Testing and QA      | Selenium, Cucumber, TDD, Postman, JUnit                                                         |
| Tools               | Git, Docker, Maven, Kafka, Tomcat                                                               |
| AI and Productivity | TensorFlow Lite, applied use of AI tools for development, debugging, documentation and learning |
| Communication       | Native Spanish, Advanced English                                                                |

### PROJECTS AND EXPERIENCES

#### Software Developer GEOCOM Uruguay 07/2022 — 01/2024

Participated in software development, maintenance, and testing for backend, web, and test automation projects. Worked on the maintenance and evolution of promotion subsystems for sales and finance applications, fixing bugs and implementing new features using Java 6 and SOAP services. Also implemented automated tests for point-of-sale systems using Java 8 and Cucumber, helping validate critical business flows.

Additionally, developed accounting and reporting microservices using Spring Framework, MongoDB, Kafka, and REST services, applying TDD practices. Also contributed to frontend development with Angular and TypeScript, and to web test automation using Java, Spring Boot, Selenium, and Cucumber.

#### Databases University Project

Designed and implemented relational database solutions. Worked with entity-relationship modeling, relational algebra, SQL, functional dependencies, normalization, transactions, concurrency, and recovery. Developed queries and analyzed schema integrity, efficiency, and consistency.

#### Computer Networks University Project

Studied the theoretical and practical foundations of the TCP/IP model, including application, transport, network, and link layer protocols. Developed a lab project using Mininet, implementing a client-server application for traffic generation and routing/forwarding functionalities for an IP router within an emulated network.

#### Software Engineering Project University Project

Continued and extended a system previously developed by another team. Added new features and developed an associated mobile application from scratch. This project involved understanding an existing codebase, adapting to previous architectural decisions, designing new functionality, and building user-oriented solutions.

#### Programming Workshop University Project

Developed a Java application using Java SE 8 and Java EE. Implemented web functionality with Servlets, JSP, and Tomcat, along with an administrator interface developed in Swing. Worked under an incremental methodology, progressively adding new features.

#### C/C++ Projects University Project

Developed applications in C and C++ focused on data structures, algorithms, object-oriented programming, design patterns, and large-scale programming concepts.

### Bilingual CV Website

### Personal Project

Developed a one-page personal website using React and Vite. Implemented smooth internal navigation, Spanish/English language switching, responsive design, project links, and a contact section. Used Tailwind CSS, lucide-react, GitHub Pages, and GitHub Actions to build and deploy a modern professional portfolio interface.

### Scoundrel Console Game

### Personal Project

Developed a Python implementation of the solitaire card game Scoundrel, executed from the console. Implemented room exploration mechanics, combat against monsters, potion usage, weapon equipment, and win/loss conditions. Designed a modular architecture separating cards, deck handling, player logic, game rules, and the console interface.

### Clear the Dungeon

### Personal Project

Developed a single-player card game in Godot 4 based on tactical dungeon clearing. Implemented monster cards, power cards, Jokers, alternative layouts, win/loss conditions, damage handling, and optional rule variants. Worked on the graphical interface, game states, turn handling, card selection logic, and Windows export.

### uconvert

### Personal Project

Developed a local Python CLI tool for file conversion and processing. Implemented features for converting images, documents, PDFs, audio/video files, ebooks, and GIS data through integration with external tools such as Pillow, Pandoc, Typst, FFmpeg, GDAL, Calibre, Ghostscript, ImageMagick, and qpdf. Added support for local installation or Docker usage, with commands for single-file conversion and batch processing.

### Smart Photo Editor

### Personal Project

Developed a web-based photo editor using canvas, designed for both desktop and mobile. Implemented canvas layers, edit history, zoom, pan, brush, eraser, shapes, text, fill, eyedropper, pixel selection, crop, filters, and export with selectable area. Integrated a React frontend with Fabric.js and an Express backend, with support for AI editing, multiple providers, visual themes, and ten languages.

### Mi Stock

### Personal Project

Developed a mobile inventory and stock management application for Android using Flutter. Implemented product creation, editing and deletion with multiple photos, barcode scanning, stock movements, history, sales, shopping cart, PDF/XLS reports, and a dashboard. Integrated SQLite, Riverpod, Material 3, internationalization, and offline visual search using TensorFlow Lite with MobileNet V3 Small.

### Download Manager

### Personal Project

Developed a desktop application in Python and PySide6 for downloading files from URL lists. Implemented link loading from `links.txt` or JSON files, manual URL entry, configurable concurrent downloads, pause, resume, retries, and SHA256 verification. Used SQLite for download history, duplicate detection, persistent configuration, and download queue organization.

### CipherChat

### Personal Project

Developed an educational prototype of an end-to-end encrypted messaging application. Implemented a React and TypeScript frontend, a Spring Boot 3 backend with Java 21, JWT authentication, PostgreSQL, Flyway, Spring Data JPA, WebSocket/STOMP, and local deployment with Docker Compose. Added support for multiple devices per user, browser-side key generation, local private-key storage in IndexedDB, encrypted per-device message delivery, direct and group chats, identity verification, and encrypted history restoration.

### HomeCinema

### Personal Project

Developed a personal LAN media server with a streaming-style interface. Implemented a Spring Boot backend, React and TypeScript frontend, SQLite database, Flyway migrations, TMDb integration, and video playback through HLS using FFmpeg. Added support for local library scanning, catalog search, playback progress tracking, HLS manifest generation, portable Windows mode, and navigation compatible with browsers and Android TV.

### DashForge 3D

### Personal Project

Developed a 3D obstacle-runner game built in C11 with OpenGL 3.3 Core. Implemented five official levels, a level editor with save/load support, custom levels, export through `.dash` files, a HUD with progress bar, and three camera modes. Used GLFW, GLAD, cglm, *stb\_image*, *stb\_asyfont*, *CMake*, and *Ninja*, with *Debug* and *Release* builds for Windows.

**Rubik's Cube 3D****Personal Project**

Developed a 3D Rubik's Cube game in Unity 6. Implemented a 3x3x3 cube with 27 individual cubies, camera rotation with middle mouse, sticker detection through raycasting, and layer turns using click and drag. Developed shuffle functionality with a timer, solve functionality through reverse playback of the move history, and automatic solved-state detection.

**EDUCATION**

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**Computer Engineering**, *Universidad de la República*

2018 - present